

Module 7 Week A — Stretch: Manual Evaluation + Calibration

Honors Track. This stretch is for learners who have completed all core assignments, are On Track or Advanced in the program, and are attending consistently.

Prerequisite: Lab 7A base must be substantially complete — you need a fine-tuned model in your local Lab 7A `model/` directory.

Due: Tuesday end of day. Resubmissions are accepted through Saturday of the assignment week.

Objective

Take the fine-tuned DistilBERT classifier from Lab 7A and evaluate it *without* `Trainer.evaluate` or any high-level metrics utility. Build a manual PyTorch inference loop, compute classification metrics from primitives, then add a calibration analysis: a reliability diagram and Expected Calibration Error (ECE).

The pedagogical goal is to expose the inference loop that `Trainer.predict` hides — and to introduce calibration as an evaluation dimension that goes beyond accuracy.

Repo and Branch

This stretch **extends your Lab 7A repo**. Create a new branch off `main`:

```
cd <your-lab-7a-repo>
git checkout main
git checkout -b stretch-tue-calibration
```

Your local `model/` from Lab 7A is exactly what you need; no cross-repo handoff.

The starter for this stretch lives in your Lab 7A template repo's `stretch/tuesday/` subdirectory. Copy those files to the repo root on this branch:

```
cp -r stretch/tuesday/* .
```

What You're Building

Files

- `manual_eval.py` — manual inference loop and metrics from primitives
- `calibration.py` — reliability diagram and ECE
- `calibration-analysis.md` — your 1-page memo
- `figures/reliability-diagram.png` — saved plot

`manual_eval.py` exposes

```
def manual_predict(model, tokenizer, texts: list[str], batch_size: int = 8) -> tuple:
    """
    Manual PyTorch inference. No Trainer.predict.

    Returns (preds, probs):
        preds: shape (N,), int class indices
        probs: shape (N, num_classes), probabilities (post-softmax)
    """

def compute_classification_report_from_arrays(y_true, y_pred) -> dict:
    """
    Compute accuracy, per-class precision/recall/F1, and macro-F1 using
    only numpy. No sklearn metric helpers, no Hugging Face evaluate.

    Returns: {
        "accuracy": float,
        "macro_f1": float,
        "per_class": {label_index: {"precision": ..., "recall": ..., "f1": ...}, ...
    }
    """
```

The point of `compute_classification_report_from_arrays` is to make the math explicit. After you've written it, the formulas should feel concrete, not magical.

`calibration.py` exposes

```
def reliability_diagram(probs: np.ndarray, y_true: np.ndarray, n_bins: int = 10):
    """
    Bin predictions by max predicted probability; for each bin, compute
    the empirical accuracy.

    Binning convention: bin edges are np.linspace(0, 1, n_bins + 1).
    A probability p falls in bin i if edges[i] <= p < edges[i+1], with
    the last bin inclusive on the right (so p == 1.0 lands in the last
    bin, not out of range).

    Returns (bucket_centers, bucket_accuracies, bucket_counts), all of length n_b
    """

def expected_calibration_error(probs: np.ndarray, y_true: np.ndarray, n_bins: int):
    """
    ECE = sum over bins of (bucket_count / N) * |bucket_accuracy - bucket_confide

    A perfectly calibrated model has ECE = 0.
    """

def plot_reliability(centers, accs, counts, output_path: str) -> None:
    """Provided in starter — saves a matplotlib reliability diagram to output_pat
```

Writing `calibration-analysis.md`

Required sections:

```
## Reliability diagram interpretation
What does your saved reliability diagram look like? Where is the model
over-confident vs. under-confident? Cite specific bucket values
(e.g., "the 0.7–0.8 confidence bucket showed accuracy of only 0.55, indicating
the model is over-confident in this range").

## Expected Calibration Error
Report your ECE. Interpret it: what does this number mean about model
trustworthiness for production use?

## A specific calibration pattern
Identify one specific pattern in the diagram (over-confidence on the majority
class, under-confidence near class boundaries, etc.) and reason about why
this pattern arose given how the model was trained.

## A proposed engineering action
What would you change in production based on these findings? Examples:
threshold-based abstention, temperature scaling for calibration, bucket-specific
re-training data collection. Be specific.
```

Peer Review (part of the grade)

After submitting your own analysis, review **one peer's** calibration analysis. Use this 4-question checklist as a comment on their PR:

- Is the reliability diagram correctly constructed (sensible buckets, accuracy axis correct)?
- Is the ECE value plausible given the diagram?
- Does the memo identify a specific calibration pattern (over-confidence on majority class, under-confidence near boundaries, etc.)?
- Does the memo propose at least one engineering action that would follow from these findings?

Your peer-review comment is part of your grade.

Submission

Open a PR from `stretch-tue-calibration` into your Lab 7A `main`. The PR is the deliverable; it does not need to be merged.

PR description includes:

- ECE value
- 1-sentence summary of the calibration pattern you found
- 1-sentence summary of the engineering action you proposed
- Link to the peer PR you reviewed

Submission instruction: Paste your PR URL into TalentLMS → Module 7 → Stretch — Tuesday (Honors) to submit this assignment.

